

PSOC™ Edge tips and tricks for security

Customer training workshop

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public

Prerequisites

- Recommended training prerequisites:
  - Introduction to PSOC™ Edge and ModusToolbox™
    - For an introduction to PSOC™ MCUs, including a getting started guide to ModusToolbox™, visit: https://www.infineon.com/product-information/psocdeveloper
    - For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, visit: PSOC™ Edge E84 Training Collection

- Development tools:
  - ModusToolbox™ v3.6
  - Eclipse IDE for ModusToolbox™ v2026.4.0
  - Edge Protect Security Suite v1.6
  - ModusToolbox™ Programming Tools v1.5.0
  - KIT_PSE84_EVAL_EPC2 BSP v1.0.0
  - Terminal Emulator
  - KIT_PSE84_EVAL

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Overview

- The following slides contain additional useful information related to security topics
- It also contains a list of all supported code examples and documentations available for the user related to PSOC™ Edge security

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Accessing peripherals from CM33 SPE

- The default code examples (for example, Hello World) have protection configurations that enable the NS domain to work correctly
- This means that by default, certain memory and peripheral regions are configured as non-secure
- These configurations are applied as a part of the cybsp_init() at the beginning of the CM33 SPE project
- After the cybsp_init(), if you try to access any non-secure peripherals or memory regions from the CM33 SPE application, it will result in a fault
- To access peripherals or memory regions from CM33 SPE, ensure that those configured as secure (this may require changes in the default memory configurations)
- One such use case that requires changes includes – accessing the SERT services from the CM33 SPE

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Booting to OSPI flash

- The PSOC™ Edge EVK includes an external Quad SPI and Octal SPI flash. By default, the extended boot is configured to boot from the QSPI flash

- Booting from OSPI flash on the PSOC™ Edge EVK requires the following policy changes to the smif_chip_select and smif_data_width policy fields. Provision the device with the changes

```
"smif_chip_select": {
                "description": "Chip select for primary external flash",
                "applicable_conf": "0, 1, 2, 3",
                "value": 0
                },
"smif_data_width": {
               "description": "Select data width used to interface the external flash",
               "applicable_conf": "4, 8",
               "value": 8
                },
```

- See the Switching between octal and quad SPI external flash memory section in Getting started AN for more details

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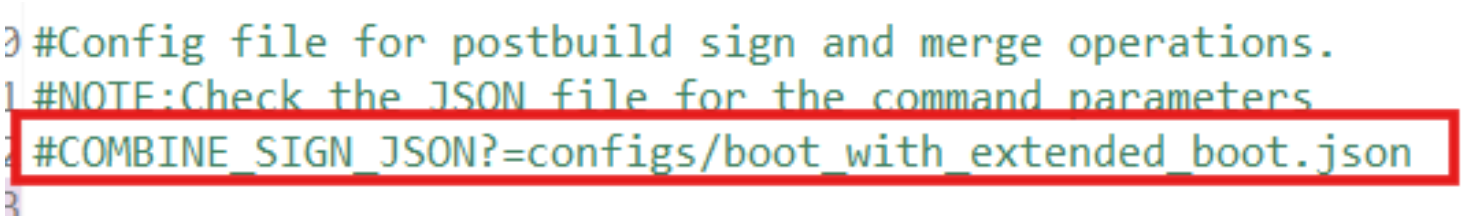
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Signing and combine user application manually

- By default, the ModusToolbox™ code examples contain configuration to sign the application after compilation
- The commands executed are determined by the COMBINE_SIGN_JSON provided in the common.mk file of the application
- To disable the signing of the application, comment or remove this line in the common.mk file



- Individual application can be signed using the Edge Protect Tools in the command line terminal (modus-shell)

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Signing and combining user application manually

EPT commands for signing applications in the Basic Secure app code example integrated with EPB.

1. $ **edgeprotecttools sign-image** --image build/project_hex/proj_bootloader.hex -s 1 --header-size 0x400 --erased-val 0 --slot-size 0x00028000 --min-erase-size 0x10 --hex-addr 0x32011000 --output build/project_hex/proj_bootloader_signed.hex --key keys/oem_private_key_0.pem
2. $ **edgeprotecttools sign-image** --image build/project_hex/proj_cm33_s.hex --header-size 0x400 --erased-val 0xFF --slot-size 0x00240000 --min-erase-size 0x10 --hex-addr 0x60100000 --protected-tlv 0x77 1 --overwrite-only --output build/project_hex/proj_cm33_s_signed.hex --key keys/oem_private_key_0.pem
3. $ **edgeprotecttools sign-image** --image build/project_hex/proj_cm33_ns.hex --header-size 0x400 --erased-val 0xFF --slot-size 0x00240000 --min-erase-size 0x10 --hex-addr 0x60340000 --protected-tlv 0x77 1 --overwrite-only --output build/project_hex/proj_cm33_ns_signed.hex --key keys/oem_private_key_0.pem
4. $ **edgeprotecttools sign-image** --image build/project_hex/proj_cm55.hex --header-size 0x400 --erased-val 0xFF --slot-size 0x00300000 --min-erase-size 0x10 --hex-addr 0x60580000 --protected-tlv 0x77 1 --overwrite-only --output build/project_hex/proj_cm55_signed.hex --key keys/oem_private_key_0.pem
5. $ **edgeprotecttools merge-hex** --image build/project_hex/proj_bootloader_signed.hex --image build/project_hex/proj_cm33_s_signed.hex --image build/project_hex/proj_cm33_ns_signed.hex --image build/project_hex/proj_cm55_signed.hex --output build/app_combined.hex

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Using alternate boot source

- Extended boot policy can be configured such that oem_app_address points to the default boot address for the user application, and oem_alt_app_address can be used to point to a fallback or alternate application

- When the user does not want the boot flow to continue to the default application, the boot flow can be changed to an alternate application (such as a firmware update application) by pulling the GPIO P17.6 to HIGH

- Setting the oem_alt_boot as false completely turns off the alternate boot feature for users who do not require this feature

***Note**: Policy changes require device provisioning.*

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Using the alternate serial interface

- Extended boot supports loading and booting of a CM33 RAM application programmed through the serial interfaces
- This process enables the application to run from the SRAM, without the need for storing any code on the flash memory
- Such RAM applications can be used for firmware update use cases

***Note**: See the code examples CM33 RAM APP for demonstration*

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Factory reset

- As long as the device is in the Development LCS, a factory-reset command can be executed to restore the device to the Infineon factory or default policy settings
- This command reverts back any custom policy configurations done by the user and also erases the OEM RoT keys provisioned by the user
- This is a useful command to be used when the device is in an unknown configuration and needs to be reset to the default
- Execute the following command in modus-shell:

```
$ edgeprotecttools -t pse8xs2 factory-reset
```


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List of security and bootloader code examples

- Basic Secure App: demonstrates secure boot and bootloader integration
- Edge Protect Bootloader: reference code example for MCUboot bootloader
- CM33 RAM App: demonstrates the alternate serial interface capability of the Extended boot
- Crypto AES: demonstration for the AES algorithm
- Crypto SHA: demonstration for the hashing algorithm
- MbedTLS PSA Crypto: shows how to use PSA APIs from the Mbed TLS library for cryptographic operations
- SRAM Loading: how to load and execute an application from SRAM using EPB
- Secure Non-secure GPIO: how to configure GPIOs as secure and non-secure
- Basic TF-M for EPC2 and EPC4: how to get started with TF-M
- TF-M Crypto for EPC2 and EPC4: how to use crypto service with TF-M
- TF-M Attestation for EPC2 and EPC4: how to use the attestation service with TF-M
- OTW update: how to perform firmware updates over serial interfaces
- DFU user-defined commands: how to define and custom DFU commands
- Secure SMIF: how to access secure SMIF peripheral from non-secure domain using SRF
- Encrypted Boot: how to perform encrypted firmware updates

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List of security and bootloader code examples (contd.)

- Power management with SRF: demonstrates secure power management with SRF
- Power management with TF-M EPC2 and EPC4: demonstrates secure power management with TF-M
- OTA BLE: demonstrates over-the-air updates via Bluetooth® interface
- OTA HTTPS: demonstrates over-the-air updates via HTTPS interface
- OTA MQTT: demonstrates over-the-air updates via MQTT interface

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Resources

- **Application notes**
  - Getting started with PSOC™ Edge security
  - PSOC™ Edge Provisioning specification
  - Edge Protect Bootloader for PSOC™ Edge
  - Device Firmware Update (DFU) middleware (MW) for ModusToolbox™
  - Getting started with Trusted Firmware-M (TF-M) on PSOC™ Edge
  - Selecting and configuring memories for power and performance in PSOC™ Edge
- **Datasheet and reference manual**
  - Architecture reference manual
  - Registers reference manual
  - Consumer datasheet
  - Industrial datasheet

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Resources (contd.)

- **Knowledge-based articles (KBA)**
  - https://community.infineon.com/t5/Knowledge-Base-Articles/Using-FreeRTOS-on-TrustZone-enabled-CM33-CPU-PSOC-Edge-E84/ta-p/1107173
  - https://community.infineon.com/t5/Knowledge-Base-Articles/Merits-and-constrains-of-Octal-DDR-memory-interface-of-PSOC-Edge-MCU/ta-p/1148428
  - https://community.infineon.com/t5/Knowledge-Base-Articles/Supported-SFDP-memories-for-PSOC-Edge/ta-p/1148429

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Revision history

Document revision	Date	Description of change
**	2026-01-18	Initial release.

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